Data Appendix: Comparing Stock Market Volatility During Different Presidential Terms

Analysis Data File: sp500_clean.csv

Unit of observation:
Each row represents a single day.

Variable: SP500

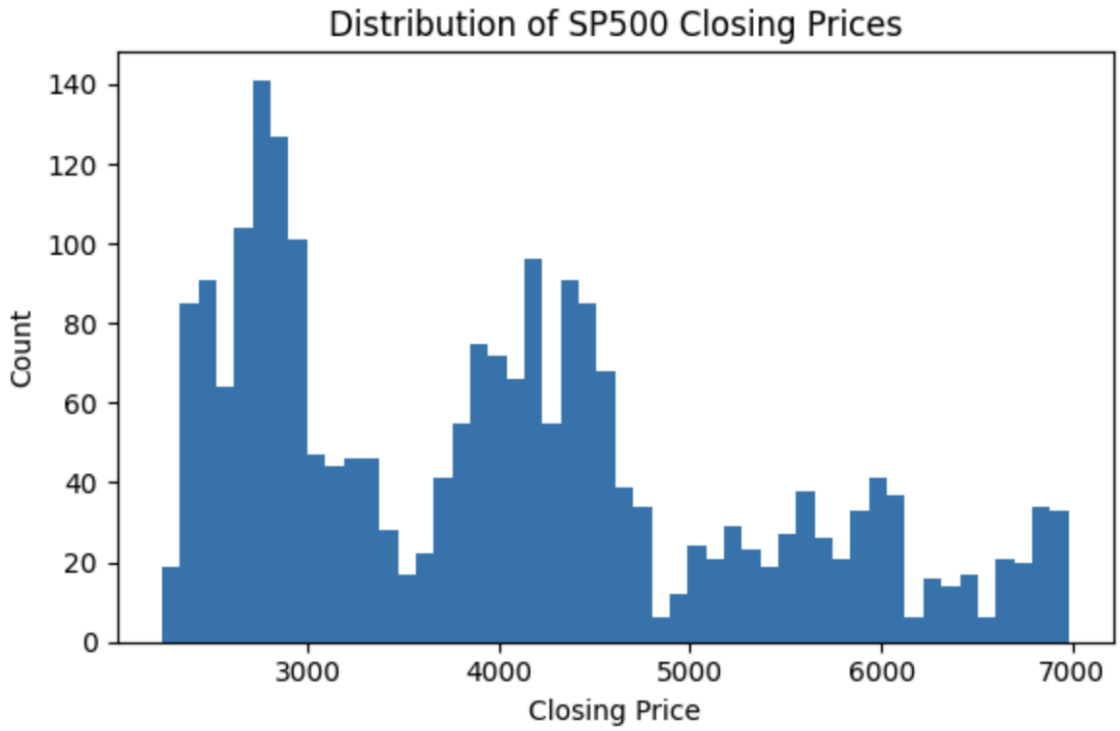
Description:
The daily index value for the S&P 500 at market close.

Type:
Decimal (float)

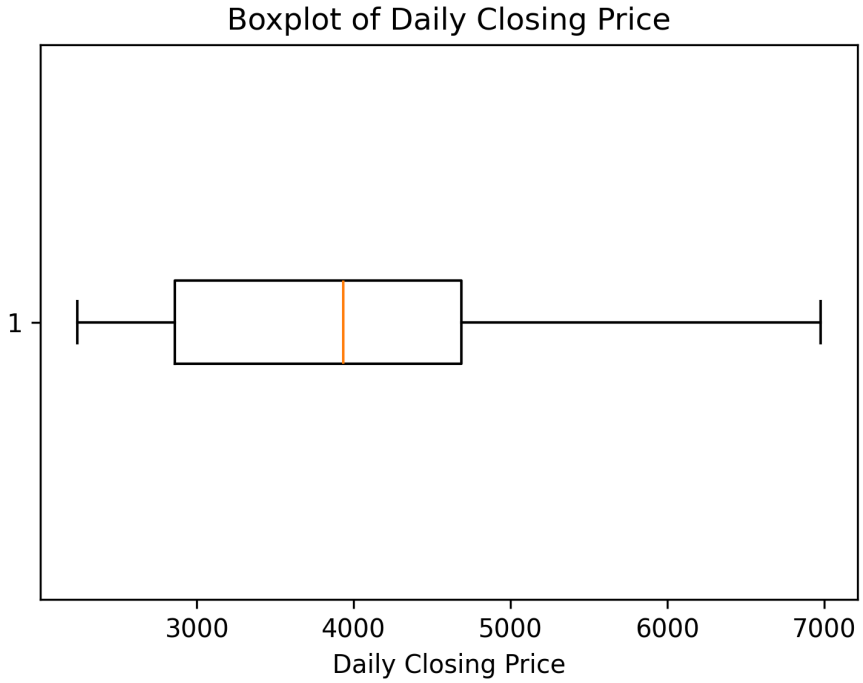
Summary statistics:
The table below reports the number of observations, the mean, median, minimum, and maximum value of S&P 500 index value at the time the market closes.

```
count    2283.000000
mean     4010.769886
std      1267.100620
min      2237.400000
25%      2858.990000
50%      3932.690000
75%      4688.675000
max      6978.600000
Name: SP500, dtype: float64
```

Distribution of Closing Prices:
The histogram shows the distribution of S&P 500 closing prices at the time of closing. The prices are spread out over a wide range with multiple different clusters, forming a right skew. This indicates that the data is coming from different time periods (the three different administrations) and that there is a difference between these three administrations.



Boxplot of headline length:
The boxplot summarizes the spread of closing prices and highlights the presence of outliers. This plot highlights the non-normal distribution of closing prices, showing its right skew.



Variable: Date

Description:
The date the index value occurred.

Type:
Datetime (object)

Summary statistics: NaNs on non-trading days

The first table below shows the amount of NaNs in the original dataset. The second table shows that these NaNs occur on non-trading days, allowing us to safely remove these rows without affecting our analysis.

```
observation_date     0
SP500               87
dtype: int64
```

	observation_date	SP500
21	2017-02-20	NaN
60	2017-04-14	NaN
91	2017-05-29	NaN
117	2017-07-04	NaN
161	2017-09-04	NaN
...	...	...
2309	2025-11-27	NaN
2329	2025-12-25	NaN
2334	2026-01-01	NaN
2346	2026-01-19	NaN
2366	2026-02-16	NaN

Variable: Returns

Description:
The difference between the current day's closing price and the previous day's closing price.

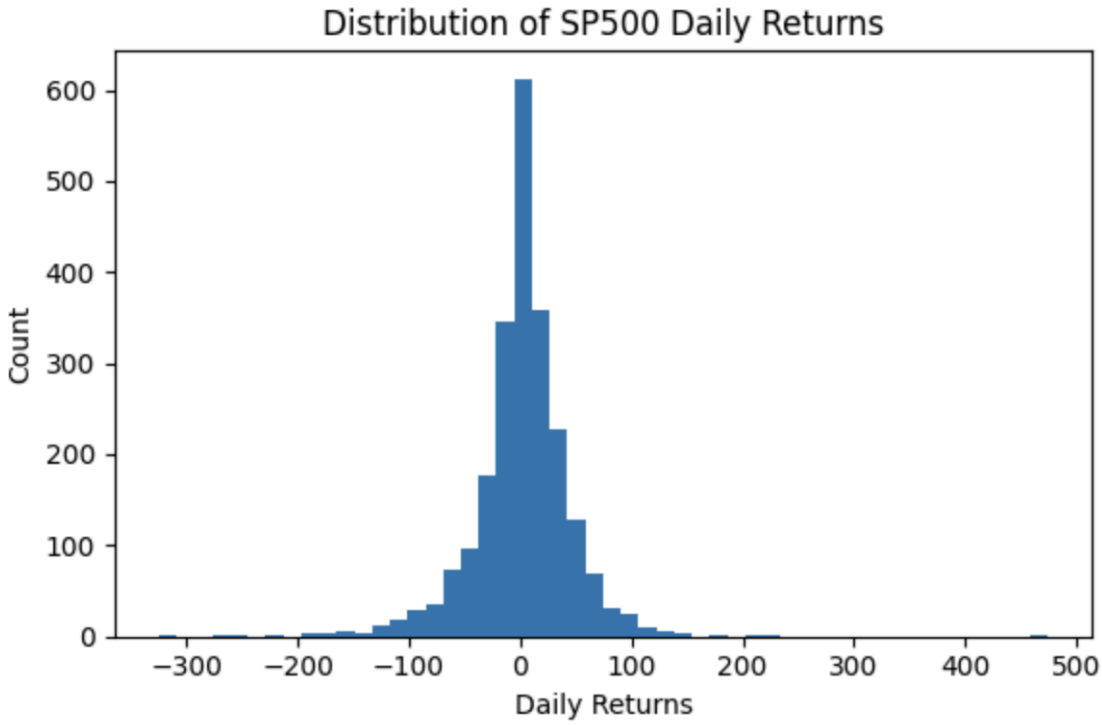
Type:
Decimal (float)

Summary Statistics:

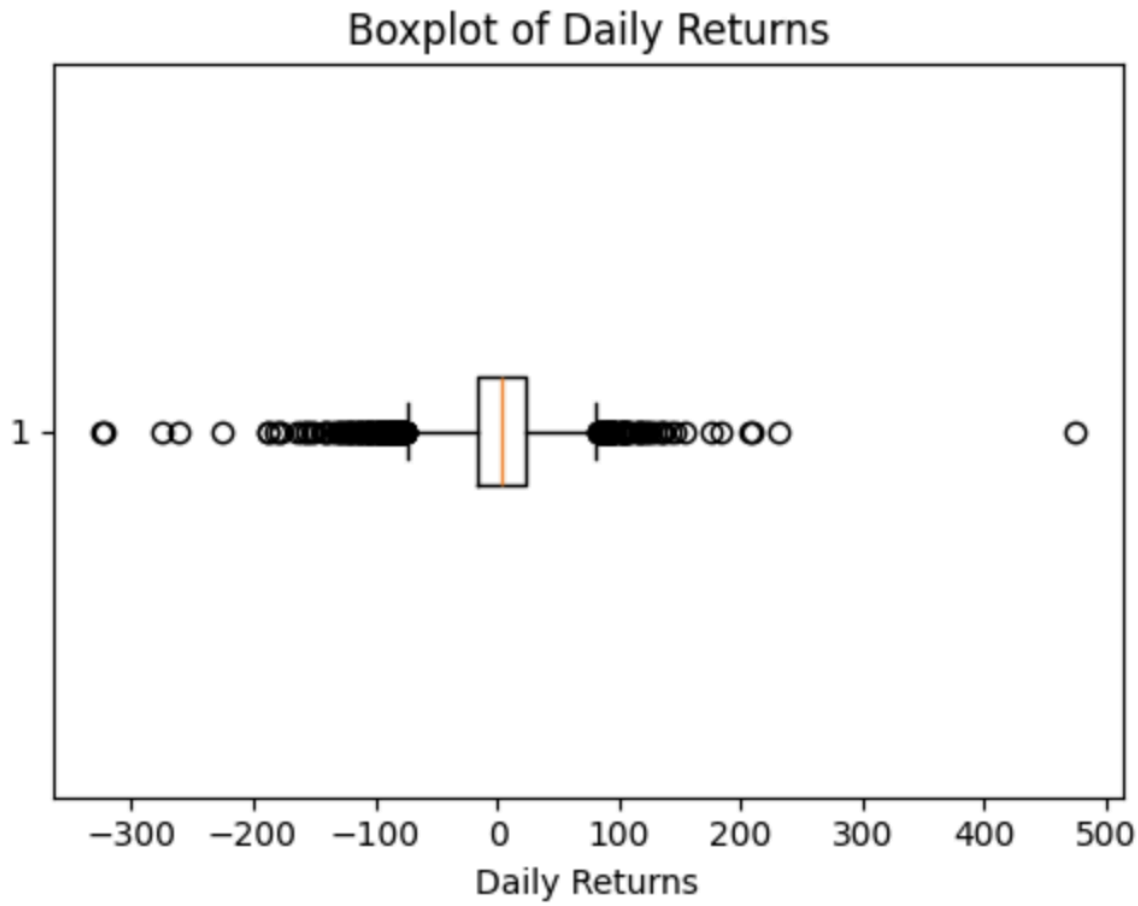
The table below reports the number of observations, the mean, median, minimum, and maximum value of returns at the time the market closes.

```
count    2283.000000
mean        2.031625
std        44.601016
min      -324.890000
25%       -15.465000
50%         2.780000
75%        23.355000
max       474.130000
Name: Returns, dtype: float64
```

Distribution of Returns:
The bar chart shows the distribution of S&P 500 daily returns. The data is normally distributed, centering around 0 which indicates that returns are relatively evenly split between positive and negative returns.



Boxplot of Retuns:
The boxplot summarizes the spread of daily returns and highlights the presence of outliers. The median return amount lies near the center of the distribution, and several outliers are visible in both directions, confirming the relatively normal distribution seen in the histogram.


Boxplot of Daily Returns
1
−300
−200
−100
0
100
200
300
400
500
Daily Returns